LLM-Text Watermarking based on Lagrange Interpolation



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1 Introduction

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- LLM-generated text can be modified, paraphrased, or partially deleted.
- Such changes can make watermark detection and authorship attribution difficult.
- **Problem:** Recover the embedded author identity from the surviving watermark, despite text modifications.

Research Gaps

- Existing watermarking methods focus largely on detection rather than recovering a specific author identity.
- Text modifications can disrupt watermark recovery.
- Distinguishing genuine watermark points from inserted or corrupted points remains challenging.
- Robust identity recovery must balance resilience against tampering with resistance to forgery.

Contributions

- Lagrange interpolation-based watermarking scheme for author identity recovery.
- Maximum Collinear Points (MCP) detection to identify genuine watermark points despite corruption.
- Extensions to multiple secrets and higher-degree polynomials.
- Experimental analysis of proposed techniques

Proposed Watermarking Scheme

- Encode the secret identity as a polynomial over $GF(2^n)$.
- Embed points $(x_i, f(x_i))$ into LLM-generated text.
- Recover watermark points from potentially modified text.
- Identify the dominant line using Maximum Collinear Points (MCP).
- Reconstruct the secret identity from the recovered line.